

EMMANUEL KUFFOUR OSEI

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GitHub: <https://github.com/ikuffourosei>

Portfolio: <https://ikuffourosei.github.io>

I am focused on addressing road-related challenges through transportation research. With proven research skills and relevant experience in GIS, Python Programming and data analysis, I am dedicated to advancing transportation engineering, aiming for solutions that enhance the safety, efficiency, and operations of transportation systems worldwide.

EDUCATION

Bachelor of Science in Geomatic Engineering

Aug 2018 – Sep 2022

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Thesis: Interactive Web Map of KNUST campus To Facilitate Navigation Technologies:

QGIS, JavaScript (Leaflet Library)

RESEARCH EXPERIENCE

Building and Roads Research Institute (BRRI) – Kumasi, Ghana

Research Assistant (Supervisor: Ing. Mrs. Emmanuella A. Asamoah)

(Nov 2022 – Oct 2023)

- Analysis of spatiotemporal distribution of traffic accidents to identify high-risk areas and times for implementing safety measures.
- Mapping and Visualization of accident data in a web map using Leaflet.
- Crash Prediction Models at Intersections of Multi-lane Highways using QGIS and Python.
- Assisted other divisions with their research by finding the research gaps in the “Affordable Housing Project in Ghana”.

Department of Urban Roads (DUR) – Kumasi, Ghana

Intern (Supervisor: Ing. Mrs. Kennyworth Baaba Buckson)

(Oct 2021 – Dec 2021)

- Assisted with the design and development of the ‘New Kumasi’ project using AutoCAD to facilitate better operations and efficiency of urban roads in Kumasi.
- Estimated the Annual Average Daily Traffic (AADT) to determine road capacities and design infrastructure that effectively accommodates the traffic volume.

PROFESSIONAL EXPERIENCE

Joshob Construction Company Limited – Kumasi Ghana
Geomatic Engineer

(Apr 2024 – Present)

Projects:

- Rehabilitation of Mankranso – Sunyani (N6) Highway (40.3km).
- Tepa College of Education: Internal Road Network (1.6km)
- Rehabilitation of Anwiankwanta – MIM Junction Road (26.4km)

Roles:

- Conduct high-precision topographic surveys and Digital Elevation Modeling (DEM) to optimize land use before project commencement.
- Supervise large-scale earthmoving operations, ensuring material efficiency.
- Compute high precision volumetric analysis of earthworks, concrete, and asphalt to ensure accurate project valuation and resource accountability.

TECHNICAL SKILLS

- **GIS and Mapping:** Google Earth Pro, QGIS, ArcGIS/ArcMap
- **Data Analysis and Prediction:** Python, Crash Data Analysis
- **Engineering and Surveying:** AutoCAD, MATLAB, GPS, Leveling, Cartography

CERTIFICATES

- **GIS FOR Climate Action (Online)-ESRI** March 2025
Key Skills: Predictive Modelling, creating Heat Index, 3D visualization of sea level rise, Optimizing delivery routes.
- **Software Engineering (Online) – ALX** Jul 2023 – Aug 2024
Key Skills: Python Programming, Data Structures, Algorithms, Software Development Lifecycle, Bash Scripting and Automation, Testing.

REFERENCES

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